

Fine particulate matter exposure and incidence of stroke

A cohort study in Hong Kong

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ABSTRACT

Objective: We aimed to assess the association of long-term residential exposure to fine particulate matter (PM) with aerodynamic diameter less than 2.5 μm (PM_{2.5}) with the incidence of stroke and its major subtypes.

Methods: We ascertained the first occurrence of emergency hospital admission for stroke in a Hong Kong Chinese cohort of 66,820 older people (65+ years) who enrolled during 1998–2001 (baseline) and were followed up to December 31, 2010. High-resolution (1 $\times$ 1 km) yearly mean concentrations of PM_{2.5} were predicted from local monitoring data and US National Aeronautics and Space Administration satellite data using linear regression. Baseline residential PM_{2.5} exposure was used as a proxy for long-term exposure. We used Cox proportional hazards to evaluate the risk of incident stroke associated with PM_{2.5} exposure adjusted for potential confounders, including individual and neighborhood factors.

Results: Over a mean follow-up of 9.4 years, we ascertained 6,733 cases of incident stroke, of which 3,526 (52.4%) were ischemic and 1,175 (17.5%) were hemorrhagic. The hazard ratio for every 10 $\mu\text{g}/\text{m}^3$ higher PM_{2.5} concentration was statistically significant at 1.21 (95% confidence interval [CI] 1.04–1.41) for ischemic and non-statistically significant at 0.90 (95% CI 0.70–1.17) for hemorrhagic stroke in fully adjusted model 3. The estimates for ischemic stroke were higher in older participants (>70 years), less educated participants, and in men for current smokers.

Conclusion: Long-term PM_{2.5} exposure was associated with higher risk of incident ischemic stroke, but the association with incident hemorrhagic stroke was less clear. *Neurology*® 2017;88:1–9

GLOSSARY

AOD = aerosol optical depth; **BMI** = body mass index; **CI** = confidence interval; **COPD** = chronic obstructive pulmonary disease; **EPD** = Environmental Protection Department; **ESCAPE** = European Study of Cohorts for Air Pollution Effects; **HR** = hazard ratio; **ICD-9** = *International Classification of Diseases-9*; **NASA** = National Aeronautics and Space Administration; **PM** = particulate matter; **PM_{2.5}** = particulate matter with aerodynamic diameter less than 2.5 μm ; **SEC** = surface extinction coefficients.

Particulate matter (PM) air pollution plays a causal role in cardiovascular morbidity and mortality, according to a definitive scientific statement from the American Heart Association.¹ Many epidemiologic studies have demonstrated higher risk of stroke associated with both short-term^{2,3} and long-term PM exposure.^{4–8} Nevertheless, stroke is heterogeneous in nature and etiology, with different risk profiles for ischemic and hemorrhagic stroke,⁹ making it important and informative to consider them separately. Few studies have examined the effects of PM air pollution on different types of stroke, with somewhat mixed results; in Western countries, the risks of air pollution are greater for ischemic than hemorrhagic stroke,^{10–12} but some studies from Asia have found that air pollution only affected hemorrhagic stroke¹³ or intracerebral hemorrhage as well as ischemic stroke.^{14,15} Few studies linking long-term PM exposure to the incidence of stroke by subtype have been conducted.^{10,16,17}

Risk factors for stroke, including hypertension, current smoking, abdominal obesity, diet, alcohol intake, and physical activity, account for more than 80% of the global risk of stroke

Supplemental data
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(ischemic and intracerebral hemorrhagic), based on a case-control study in 22 countries.¹⁸ Air pollution has been proposed as an emerging global risk factor for stroke.⁹ In this study, in a population with low levels of smoking and alcohol use but high levels of exposure to PM with aerodynamic diameter less than 2.5 μm (PM_{2.5}),¹⁹ we assessed the association of long-term residential exposure to PM_{2.5} with the incidence of overall, ischemic, and hemorrhagic stroke. Hong Kong is an ideal place to study these associations because hemorrhagic stroke is relatively common²⁰ and high levels of public housing mean less residential segregation,²¹ and thereby less confounding by socioeconomic position. We also assessed whether the associations varied by age, sex, socioeconomic position, or smoking to identify vulnerable subpopulations.

METHODS Study population. A total of 66,820 adults aged 65 years or older were enrolled by the Elderly Health Service of the Department of Health in Hong Kong from July 1998 to December 2001. Elderly Health Centres located in each of the 18 districts in Hong Kong provide health assessments, using standardized and structured interviews, and comprehensive clinical examinations. Information on sociodemographics, lifestyle, and disease history was collected by doctors and nurses, as described previously.¹⁹ The participants account for about 9% of the older people in Hong Kong.¹⁹ The cohort was set up to promote understanding of aging in this developed non-Western setting where the patterns of common chronic diseases and their determinants may differ from those in the West.

Standard protocol approvals, registrations, and patient consents. The protocol was approved by the Institutional Review Board of the University of Hong Kong/Hospital Authority Hong Kong West Cluster and the ethics committee of the Department of Health. This is an analysis of routinely collected data; the participants implicitly agreed to their data being used for research by using the service.

Exposure model and estimation of residential exposure to PM_{2.5}. We assessed PM_{2.5} exposure from satellite-based aerosol optical depth (AOD) recordings and monitoring data from ground-based stations.²² AOD, a measure indicating PM_{2.5} levels in the troposphere, was retrieved for Hong Kong from remote sensing imaging by National Aeronautics and Space Administration (NASA)'s 2 Earth Observing System satellites.²³ We estimated surface extinction coefficients (SEC) from AOD at 1 $\times$ 1 km resolution, after controlling for humidity and rainy days.²⁴ The missing SEC data (15.7%) mainly due to cloud cover problems were replaced by a multiple imputation procedure.²² Four general monitoring stations, maintained by the Environmental Protection Department (EPD) in Hong Kong, continuously monitor PM_{2.5} concentrations. Annual mean PM_{2.5} concentrations from 1998 to 2010 for each general monitoring station were averaged from hourly concentrations and then regressed on annual SEC. For each year between 1998 and 2010, annual PM_{2.5} exposures at the residential location of each

participant, based on geocoded anonymized address, were obtained using the same exposure model with annual SEC as the explanatory variable. In this Hong Kong cohort of older people, about 13.3% changed their residential address during the period 1998 to 2010. We used PM_{2.5} exposure in the year of recruitment for each participant to approximate long-term exposure in the main analyses.²² PM_{2.5} exposure year by year as a time-varying variable was used in a sensitivity analysis,⁴ taking into account changes in residential address.

Health endpoints: Incidence of stroke and its subtypes.

We obtained stroke incidence by subtype from 1998 to 2010 from hospitalization records via record linkage to the electronic health record system of the Hospital Authority, which manages all 42 public hospitals in the entire territory of Hong Kong. All hospital discharges are coded using the ICD-9: 430–436,²⁵ with hemorrhagic stroke defined as ICD-9 430, 431, ischemic stroke as ICD-9 433, 434, and unspecified stroke as ICD-9 436. We ascertained the first occurrence of emergency hospital admissions (admissions through the accident and emergency services) for stroke and its major subtypes as the incident cases.

Individual, ecological, and environmental covariates.

We identified potential confounders from the literature as possible common causes of PM_{2.5} exposure and stroke. On this basis, age, sex, body mass index (BMI), physical exercise, and socioeconomic position (education and personal monthly expenditure) were considered as potential confounders, because these might result in residence in a location more prone to air pollution. To characterize differences between residential areas more fully, we also included neighborhood tertiary education, monthly domestic household income, and percentage of older people (aged 65+) based on 197 small areas from the 2001 census, called tertiary planning units. We also included the percentage of smokers (aged 15+) in the 18 districts of Hong Kong to indicate the exposure to environmental tobacco smoke, as previous studies suggest.^{22,26} We also collected information on active chronic diseases and medication use at baseline. We defined self-reported active diseases as self-reported hypertension, heart disease, diabetes, chronic obstructive pulmonary disease (COPD)/asthma, or cerebrovascular accident. We used 3 models to estimate the hazard ratios (HR) for every 10 $\mu\text{g}/\text{m}^3$ higher PM_{2.5} concentration adjusted for different sets of potential confounders. Model 1 was adjusted for age and sex. Model 2 was additionally adjusted for BMI, physical exercise, education, monthly expenditure, and self-reported active diseases. Model 3 was additionally adjusted for smoking status, alcohol use, medication use, and neighborhood level factors including the percentage of older people, the proportion with tertiary education, and proportion with income $\geq$ US \$1,923/mo, and the smoking rate at the district level. Model 3 does not include self-reported active diseases, such as hypertension and diabetes, as confounders because these could be mediators of the associations of air pollution with stroke. All confounders were included in the models as the baseline values considered as time-independent variables.⁸

Statistical analysis. We used Cox proportional hazards to estimate the adjusted association of long-term exposure to PM_{2.5} with incidence of stroke overall and by subtype.⁴ Follow-up time from baseline (time to event) was used as the timescale, which was from date of recruitment to date of first hospital admission for stroke, date of death, or censored at the end of follow-up on December 31, 2010. We used the estimated PM_{2.5} exposure at baseline (enrollment from 1998 to 2001) as a time-independent variable to represent long-term exposure in the main analyses.²² The Cox proportional hazard assumption was tested using the

cox.zph function in the survival package in R. The exposure-response relation of PM_{2.5} with stroke overall and by subtype was plotted using a natural cubic spline with 3 degrees of freedom for PM_{2.5} exposure term in model 3.²² Linearity was tested by comparing the fit of the linear and the spline model using the log likelihood ratio χ^2 test.²⁷

To assess potential effect modification, we conducted subgroup analyses stratified by age (≤ 70 or > 70 years), sex (male or female), education level (below primary, primary, or secondary and above), and smoking status (never smoker, former smoker, or current smoker). The *p* value for the interaction was obtained from the interaction of each potential effect modifier with PM_{2.5} in model 3.

To assure detection of long-term associations and to control for competing diseases, we performed sensitivity analyses excluding participants with incident stroke or death in the first year after entry, or excluding those with self-reported cerebrovascular accident at baseline.²² As air pollution exposure may be related to exposures across the life course, we also conducted a sensitivity analysis with attained age as the underlying timescale while adjusting for calendar year of enrollment. Using attained age as the time scale provides flexible control for age while avoiding the need to include age as a confounder.²⁸ Considering PM_{2.5} exposure may vary from year to year, a time-dependent model for yearly PM_{2.5} exposure from 1998 to 2010 was used in sensitivity analysis.⁴

All analyses were conducted in R statistical environment version 3.3.0, with survival package for survival analysis to estimate the hazard ratio, rms for regression modeling strategies, and Hmisc for plotting the exposure-response relationship curve.

RESULTS A total of 66,820 older people were enrolled in the initial study cohort. After excluding 1,932 (2.9%) people without sufficient address

information for geocoding, 3,420 (5.1%) with incorrect geocoding or without PM_{2.5} exposure estimates due to lack of satellite data, and 21 (0.03%) with missing covariates, a final sample of 61,447 (92.0%) was included in the analyses.

The correlation between the annual PM_{2.5} concentrations obtained from the general monitoring stations and from the adjusted satellite-based SEC on dry days in the same 1×1 km grid during 1998–2010 was 0.625 (*p* value < 0.001). The mean estimated annual PM_{2.5} concentration in the baseline year was 35.8 $\mu\text{g}/\text{m}^3$ with an SD of 2.4 $\mu\text{g}/\text{m}^3$.

The spatial distribution of the 61,447 participants included in this study is shown in figure 1. Over a mean follow-up of 9.4 years (a total of 578,464.7 person-years), we observed 6,733 cases of incidence stroke with an incident rate of 11.6 cases per 1,000 person-years in this cohort of older people from Hong Kong. Most strokes (52.4%) were ischemic, 26.5% were unspecified, and 17.5% were hemorrhagic.

In this cohort of older people, the mean age at entry was 72 years and about 65.9% were women. A total of 5,907 (9.6%) were current smokers and 11,871 (19.3%) were former smokers. A total of 8,461 (13.8%) were former or regular drinkers and 44,252 (72%) did daily physical exercise. Most participants (82.9%) had primary school education or below. About half the participants took regular medication (53.1%) and reported active diseases (48.9%), including hypertension, heart disease,

Figure 1 Spatial distribution of the patients in the Elderly Health Services cohort in Hong Kong (n = 61,447) at baseline (1998–2000) by stroke occurrence

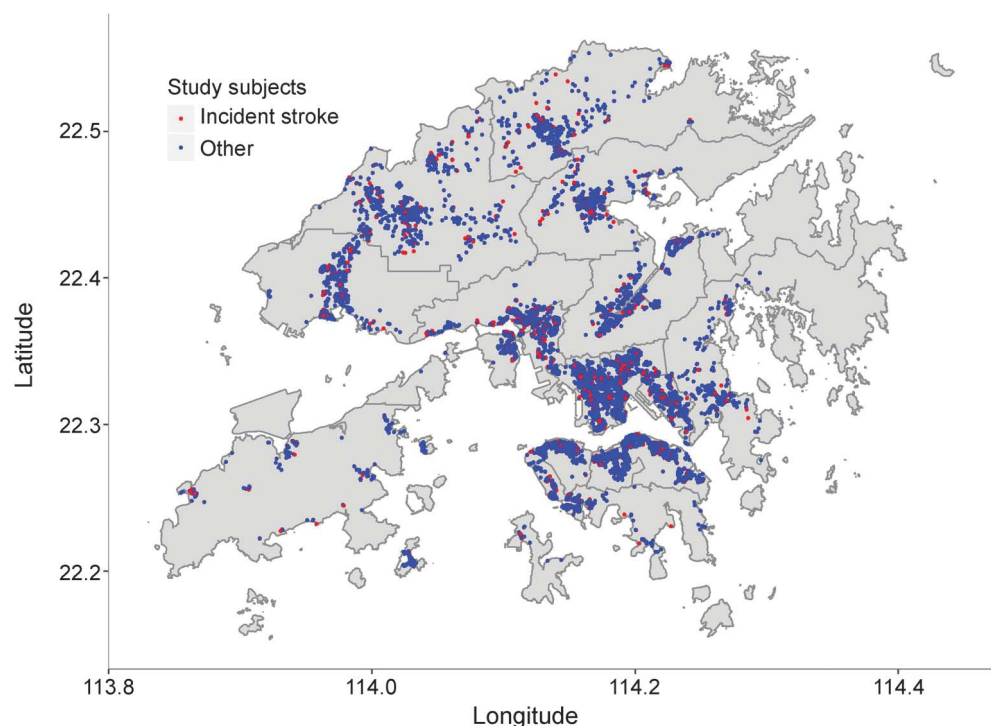


Table 1 Descriptive characteristics of the Elderly Health Services cohort (n = 61,447) at baseline (1998–2000) by particulate matter with aerodynamic diameter less than 2.5 μm (PM_{2.5}) exposure ($\mu\text{g}/\text{m}^3$)

Characteristic	All patients (n = 61,447)	PM _{2.5} exposure ^a	
		Low (n = 30,828)	High (n = 30,619)
PM _{2.5} concentrations, $\mu\text{g}/\text{m}^3$, mean $\pm$ SD	35.8 $\pm$ 2.4	33.9 $\pm$ 1.5	37.7 $\pm$ 1.5
Incident stroke, n (%)	6,733 (11.0)	3,269 (10.6)	3,464 (11.3)
Hemorrhagic	1,175 (1.9)	593 (1.9)	582 (1.9)
Ischemic	3,526 (5.7)	1,699 (5.5)	1,827 (6.0)
Unspecific	1,785 (2.9)	860 (2.8)	925 (3.0)
Individual-level covariates			
Age at entry, y, n (%)			
≤ 70	29,038 (47.3)	14,801 (48.0)	14,237 (46.5)
> 70	32,409 (52.7)	16,027 (52.0)	16,382 (53.5)
Sex, n (%)			
Male	20,933 (34.1)	10,579 (34.3)	10,354 (33.8)
Female	40,514 (65.9)	20,249 (65.7)	20,265 (66.2)
BMI quartiles, n (%)			
1st (< 21.6)	14,168 (23.1)	6,948 (22.5)	7,220 (23.6)
2nd–3rd (21.6–26.3)	31,179 (50.7)	15,529 (50.4)	15,650 (51.1)
4th (> 26.3)	16,100 (26.2)	8,351 (27.1)	7,749 (25.3)
Smoking status, n (%)			
Never	43,669 (71.1)	22,031 (71.5)	21,638 (70.7)
Former	11,871 (19.3)	5,896 (19.1)	5,975 (19.5)
Current	5,907 (9.6)	2,901 (9.4)	3,006 (9.8)
Alcohol drinking, n (%)			
Never/social drinker	52,986 (86.2)	26,597 (86.3)	26,389 (86.2)
Former/regular drinker	8,461 (13.8)	4,231 (13.7)	4,230 (13.8)
Exercise in d/wk, n (%)			
Never (0)	9,406 (15.3)	4,335 (14.1)	5,071 (16.6)
Medium (1–6)	7,789 (12.7)	3,862 (12.5)	3,927 (12.8)
High (7)	44,252 (72.0)	22,631 (73.4)	21,621 (70.6)
Education, n (%)			
Below primary	28,242 (46.0)	14,455 (46.9)	13,787 (45.0)
Primary	22,656 (36.9)	10,967 (35.6)	11,689 (38.2)
Secondary or above	10,549 (17.2)	5,406 (17.5)	5,143 (16.8)
Expenses/mo in USD, n (%)			
Low (< 128)	10,122 (16.5)	4,338 (14.1)	5,784 (18.9)
Medium (128–384)	42,152 (68.6)	21,554 (69.9)	20,598 (67.3)
High (≥ 385)	9,173 (14.9)	4,936 (16.0)	4,237 (13.8)
Medication taken, n (%)			
Yes	32,628 (53.1)	14,518 (47.1)	14,301 (46.7)
No	28,819 (46.9)	16,310 (52.9)	16,318 (53.3)
Active diseases, ^b n (%)			
Yes	30,052 (48.9)	15,903 (51.6)	15,492 (50.6)
No	31,395 (51.1)	14,925 (48.4)	15,127 (49.4)

Continued

Table 1 Continued

		PM _{2.5} exposure ^a	
Characteristic	All patients (n = 61,447)	Low (n = 30,828)	High (n = 30,619)
TPU level covariates			
Prevalence of age ≥65 y, mean	12.10	11.38	12.84
Prevalence of tertiary education, mean	12.98	13.38	12.59
Prevalence of income ≥\$1,923/mo USD, mean	59.50	61.93	57.04
District level covariate			
Smoking rate, mean	11.55	11.49	11.61

Abbreviations: BMI = body mass index; TPU = tertiary planning units.

^aHigh and low PM_{2.5} exposure was defined by using the median (35.7756 μg/m³) as the cutoff point.

^bActive diseases were defined as self-reported hypertension, heart diseases, diabetes, chronic obstructive pulmonary disease/asthma, or cerebrovascular accident at baseline.

diabetes, COPD/asthma, or a cerebrovascular accident (table 1).

Effects of PM air pollution. In all 3 models, PM_{2.5} was clearly associated with higher stroke incidence (table 2) with fairly similar estimates. The HR for every 10 μg/m³ higher PM_{2.5} concentration was 1.14 (95%

confidence interval [CI] 1.02–1.27) for all incident stroke in model 3. Long-term PM_{2.5} exposure had different associations by stroke subtype, with a statistically significant HR (1.21, 95% CI 1.04–1.41) for ischemic stroke and no significant association for hemorrhagic stroke (HR 0.90, 95% CI 0.70–1.17). PM_{2.5} was also strongly associated with unspecified stroke. Sensitivity analysis excluding participants with incident stroke or death in the first year after entry, a self-reported cerebrovascular accident at baseline, or using attained age as the underlying timescale gave similar findings (table 3). However, using a time-dependent exposure of yearly PM_{2.5} from 1998 to 2010 gave less consistent estimates by stroke subtype, although the estimate for ischemic stroke remained similar (table 3). Comparison of the linear and spline models suggested the exposure–response relationship was essentially linear for all incident stroke, ischemic stroke, and unspecified stroke subtypes but not for hemorrhagic stroke (figure e-1 at Neurology.org).

Effect modification for long-term PM_{2.5} exposure. Table 4 shows the associations of PM_{2.5} exposure with incident stroke and its subtypes in model 3 stratified by age, sex, education, and smoking status. The association of PM_{2.5} with stroke varied by age and the association was only evident for those aged >70 years. Some of the other associations varied by strata but there was no other evidence of differences by strata (*p* values for interaction >0.05). The point estimates for ischemic stroke appeared higher in older people, less educated people, and in men for current smokers albeit with wide CIs.

DISCUSSION In an understudied non-Western population exposed to high levels of air pollution, we found higher risks of incident stroke, ischemic stroke, and unspecified stroke (mostly likely ischemic stroke²⁹) associated with long-term PM_{2.5} exposure. This finding is consistent with a previous meta-

Table 2 Association of 10 μg/m³ of particulate matter with aerodynamic diameter less than 2.5 μm (PM_{2.5}) at baseline (1998–2000), i.e., time-independent, with incident stroke (time to first emergency hospitalization) from 1998 to 2012, overall and by subtype, in the Elderly Health Services cohort (n = 61,447)

Incidence cases	No.	Hazard ratio	p Value
Model 1			
Stroke	6,733	1.17 (1.06–1.29)	0.002
Hemorrhagic stroke	1,175	0.99 (0.78–1.26)	0.927
Ischemic stroke	3,526	1.19 (1.04–1.37)	0.012
Unspecified stroke	1,785	1.40 (1.15–1.70)	<0.001
Model 2			
Stroke	6,733	1.18 (1.07–1.31)	0.001
Hemorrhagic stroke	1,175	0.97 (0.77–1.24)	0.833
Ischemic Stroke	3,526	1.22 (1.06–1.40)	0.006
Unspecified stroke	1,785	1.44 (1.19–1.75)	<0.001
Model 3			
Stroke	6,733	1.14 (1.02–1.27)	0.020
Hemorrhagic stroke	1,175	0.90 (0.70–1.17)	0.448
Ischemic stroke	3,526	1.21 (1.04–1.41)	0.012
Unspecified stroke	1,785	1.31 (1.05–1.62)	0.015

Model 1 adjusted for age and sex. Model 2 adjusted for what we consider to be confounders: body mass index, physical exercise, education, monthly expenses, and self-reported active diseases including hypertension, heart diseases, diabetes, chronic obstructive pulmonary disease/asthma, or cerebrovascular accident at baseline. Model 3 adjusted for confounders in model 2 except self-reported active diseases, and other studies consider to be confounders: smoking status, alcohol drinking, medication taken, the tertiary planning units level covariates (prevalence of age ≥65 years, tertiary education, and income ≥\$1,923/mo USD), and smoking rate at district level; all covariates were included in the model as fixed values at baseline.

Table 3 Sensitivity analyses in model 3 for the association of 10 $\mu\text{g}/\text{m}^3$ of particulate matter with aerodynamic diameter less than 2.5 μm ($\text{PM}_{2.5}$) at baseline (1998–2000) with incident stroke (time to first emergency hospitalization) from 1998 to 2012, overall and by subtype, in the Elderly Health Services cohort (n = 61,447)

Sensitivity analysis	No.	HR (95% CI)	p Value
Excluding those with incident stroke (n = 630) or death (n = 230) in the first year			
All stroke	6,103	1.15 (1.02–1.29)	0.019
Hemorrhagic	1,070	0.91 (0.69–1.20)	0.489
Ischemic	3,326	1.22 (1.04–1.42)	0.012
Unspecified	1,418	1.34 (1.05–1.70)	0.017
Excluding those with self-reported cerebrovascular accident at baseline (n = 1,817)			
All stroke	6,244	1.12 (1.00–1.26)	0.044
Hemorrhagic	1,089	0.87 (0.66–1.14)	0.314
Ischemic	3,274	1.20 (1.03–1.41)	0.019
Unspecified	1,616	1.28 (1.02–1.60)	0.031
With attained age rather than time to event as the underlying timescale			
All stroke	6,733	1.11 (1.00–1.24)	0.060
Hemorrhagic	1,175	0.89 (0.68–1.15)	0.365
Ischemic	3,526	1.19 (1.03–1.38)	0.022
Unspecified	1,785	1.24 (1.00–1.54)	0.045
With yearly $\text{PM}_{2.5}$ exposure as a time-varying exposure from 1998 to 2010			
All stroke	6,723	1.20 (1.06–1.37)	0.005
Hemorrhagic	1,175	1.22 (0.90–1.65)	0.210
Ischemic	3,516	1.19 (1.00–1.41)	0.050
Unspecified	1,785	1.25 (0.96–1.62)	0.095

Abbreviations: CI = confidence interval; HR = hazard ratio.

analysis of 11 cohorts in the European Study of Cohorts for Air Pollution Effects (ESCAPE), which suggested a 19% higher risk of incident stroke per 5 $\mu\text{g}/\text{m}^3$ increment in $\text{PM}_{2.5}$ exposure.⁸ Our study also shows that the exposure–response relationships were essentially linear for overall, ischemic, and unspecified stroke, and that the association of $\text{PM}_{2.5}$ exposure with hemorrhagic stroke was less clear.

While some previous studies have linked $\text{PM}_{2.5}$ to a higher risk of incident stroke,^{6–8} the evidence is not always consistent, as in the all-male Health Professionals cohort¹⁷ and some other cohorts from the United States⁴ and Sweden.⁵ These differences may have arisen from different exposure assessment models and exposure windows, different population characteristics, or different atmospheric environments with different air pollution characteristics and weather factors. We also found that long-term $\text{PM}_{2.5}$ exposure was associated with a higher risk of ischemic but not clearly with hemorrhagic stroke, consistent with a large cohort study from Denmark¹¹ and a small-area level ecological study from London (United Kingdom).¹⁰ A Canadian study observed associations of long-term $\text{PM}_{2.5}$ exposure with both hemorrhagic and nonhemorrhagic stroke subtypes¹⁶; however, the

estimates of risk diminished after controlling for ecological measures of income and deprivation.

The biological mechanisms linking long-term PM exposure with chronic damage to the cerebrovascular system may occur through several pathways. Particulate matter might increase blood pressure through disrupting hemodynamic balance favoring vasoconstriction and augmenting release of various pro-oxidative, inflammatory mediators.³⁰ However, this would suggest, if anything, a greater effect on hemorrhagic than ischemic stroke. Long-term exposure to PM may induce acceleration of atherosclerosis, alter vasomotor tone, and cause vascular inflammation,^{31,32} which might be more specific to ischemic stroke. Finally, $\text{PM}_{2.5}$ may promote blood coagulation,³³ which would be more likely to provoke ischemic than hemorrhagic stroke. However, the mechanisms underlying the differential effects of long-term $\text{PM}_{2.5}$ exposure on ischemic and hemorrhagic stroke subtypes is still not clear and needs to be further studied.

The effect of long-term $\text{PM}_{2.5}$ exposure tended to be greater at older ages and appeared to have higher point estimates for less educated people and in men for current smokers. Greater effects of PM on mortality and morbidity have previously been observed in

Table 4 Association of 10 $\mu\text{g}/\text{m}^3$ of particulate matter with aerodynamic diameter less than 2.5 μm (PM_{2.5}) at baseline (1998–2000), i.e., time-independent, with incident stroke (time to first emergency hospitalization) from 1998 to 2012, overall and by subtype, stratified by age, sex, smoking status, and education in the Elderly Health Services cohort (n = 61,447) using model 3

Stratified characteristic	All stroke		Hemorrhagic stroke		Ischemic stroke	
	No.	HR (95% CI)	No.	HR (95% CI)	No.	HR (95% CI)
Age, y						
≤70	2,334	0.97 (0.81–1.17)	419	0.94 (0.60–1.45)	1,242	1.14 (0.89–1.47)
>70	4,399	1.23 (1.08–1.41) ^{a,b}	756	0.88 (0.64–1.22)	2,284	1.25 (1.04–1.50) ^b
Sex						
Male	2,485	1.15 (0.96–1.37)	424	1.24 (0.81–1.91)	1,256	1.28 (1.00–1.64) ^b
Female	4,248	1.14 (0.99–1.31)	751	0.76 (0.55–1.06)	2,270	1.18 (0.98–1.43)
Education						
Below primary	3,291	1.21 (1.04–1.42) ^b	581	0.96 (0.66–1.39)	1,742	1.34 (1.09–1.66) ^b
Primary	2,438	1.08 (0.90–1.29)	427	0.94 (0.61–1.46)	1,265	1.05 (0.82–1.36)
Secondary or above	1,004	1.09 (0.82–1.44)	167	0.69 (0.35–1.38)	519	1.25 (0.85–1.84)
Smoking status in men						
Never smoker	906	1.10 (0.82–1.47)	153	1.02 (0.50–2.06)	452	1.28 (0.85–1.94)
Former smoker	1,046	1.14 (0.87–1.51)	172	1.36 (0.69–2.68)	523	1.07 (0.72–1.58)
Current smoker	533	1.25 (0.85–1.84)	99	1.50 (0.61–3.69)	281	1.83 (1.07–3.13) ^b
Smoking status in women						
Never smoker	3,675	1.12 (0.96–1.29)	628	0.80 (0.56–1.15)	1,999	1.16 (0.95–1.41)
Former smoker	394	1.31 (0.83–2.05)	76	0.50 (0.17–1.42)	199	1.34 (0.71–2.54)
Current smoker	179	1.40 (0.73–2.71)	47	0.86 (0.23–3.13)	72	1.72 (0.62–4.78)

Abbreviations: CI = confidence interval; HR = hazard ratio.

^ap Value for interaction between PM_{2.5} exposure and age group is 0.047 for all stroke, while p values for the other interactions in this table are all >0.05.

^bStatistically significant HR.

the elderly.³⁴ People with lower education may have to live in more polluted areas, generating greater vulnerability to PM exposure.³⁵ The stronger association in current smokers suggests a synergistic effect of long-term PM_{2.5} exposure and smoking on stroke, which has been observed before.^{11,36} Baseline smoking rates for this elderly cohort were 20.5% in men and 4.0% in women. Lifetime smoking exposure raises the risk of carotid atherosclerosis,³⁷ so smokers may be more susceptible to the effects of PM_{2.5} exposure. However, in ESCAPE, based on 11 European cohorts, associations with long-term PM_{2.5} exposure were only evident among never-smokers.⁸ Differences between studies may relate to varying demographic characteristics, smoking rates, pollution levels, and ascertainment of stroke outcomes in these cohorts.^{8,9}

We took advantage of a very large cohort, where we were able to control for a number of individual and neighborhood level confounders; nevertheless, limitations exist. First, residual confounding is possible. For example, we did not have information on traffic noise or fast food restaurant density, which have been linked to both PM_{2.5} exposure and

stroke.^{38,39} Second, we obtained hospitalization records for the participants from 1998 to 2010 and used the first admission for stroke during this period as the incident stroke. Hospitalization records prior to 1998 are not available so we could not identify participants who had previously had a stroke, which may distort the association we found. Sensitivity analysis excluding the participants with incident stroke or death in the first year after entry, or excluding those with a self-reported cerebrovascular accident at baseline, gave similar estimates. Third, the participants were enrolled at a preventive service, so they may be more health-conscious and perhaps less susceptible to the effects of air pollution than the general elderly population,¹⁹ which might make our estimates conservative. Fourth, hemorrhagic stroke is less common than ischemic stroke, and was only 17.5% of strokes, giving less power to detect the risk of this type of stroke. Fifth, we used exposure to PM_{2.5} at the residential location to approximate exposure to PM_{2.5}, which previous studies have suggested is a valid proxy for long-term exposure.^{7,40} Results were similar for all stroke, ischemic stroke, and unspecified stroke using

time-varying exposures, although the point estimate for hemorrhagic stroke was somewhat different but in both analyses indicated no clear association. Sixth, baseline residential exposure to PM_{2.5} pollution was estimated from the relation between local monitoring data measured by the Hong Kong EPD and SEC data from NASA satellites. About 15.7% of SEC data was missing due to cloud cover, which may bias the exposure assessment, although we imputed such SEC data using multiple imputation.²² Seventh, short-term PM_{2.5} changes are an established risk factor for stroke,^{2,3} so the effects of long-term PM_{2.5} exposure could be due to its correlation with short-term PM_{2.5} changes. Such confounding is unlikely because long-term exposure contrasts are established geographically in cohort studies while the short-term exposure contrasts are usually constructed temporally in time-series studies. Finally, the data used to verify the estimation model for PM_{2.5} was from 4 monitoring stations. Further study with PM_{2.5} data measured at a larger number of monitoring stations or using different approaches for exposure modeling could improve the accuracy of the exposure assessment and help confirm the health effects of long-term PM exposure on stroke and its subtypes.

This study adds to the evidence base that long-term residential PM_{2.5} exposure increases the risk of incident ischemic stroke in older people. The association of PM_{2.5} with hemorrhagic stroke was less clear.

AUTHOR CONTRIBUTIONS

H.Q. and L.T. defined the research theme and wrote the manuscript. H.Q. and S.S. analyzed the data and interpreted the results. H.T. worked on data cleaning and preliminary analyses. C.-M.W. set up the method for exposure assessment. R.S.-y.L. and C.M.S. set up the cohort. C.M.S. and L.T. reviewed/revised the manuscript and approved the submission.

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